

SV-16 microcontroller board — PCB component specifications for KiCad

What this is: the component list, the specifications and the pin-level connections needed to draw the SV-16 microcontroller board in KiCad and order it. Every part here is the one finalised in `BOARD.md` revision 1.1 (the India-sourced edition) — same parts, same values, plus the mechanical and electrical detail a schematic and layout need.

Sources of truth, in order:

Document	What it fixes
<code>constraints/ecp5_144tqfp.lpf</code>	the FPGA pin map — the §5 table in this document is generated from it, not copied
<code>BOARD.md</code> §3	the finalised BOM (India-sourced) and prices
<code>BOARD.md</code> §4, §7	the power tree and the five connection-diagram sheets
<code>BOARD.md</code> §6	which FPGA pin is on which board net
<code>BOARD.md</code> §11	layout and EMC rules, restated as KiCad rules in §7 here
<code>docs/</code> <code>SYNTHESIS_AND_DEPLOYMENT.md</code>	how the bitstream that matches this pin map is built

Two reading rules before you start drawing:

- Rows marked ⚠ must be confirmed against the part's datasheet before you commit the schematic.** They are values where the board document gives the intent but the datasheet gives the number — regulator compensation, the feedback divider, footprint sizes that depend on what you buy. §10 lists them in one place.
- Footprint names are KiCad 7/8 standard-library names.** The library prefix (`Package_QFP:` and so on) is the library nickname in KiCad's Footprint Library table. Where a dimension depends on the exact part you buy (electrolytics, inductors, USB-C, oscillator, module headers), the row says so.

1. Setting up the KiCad project

Step	Setting
KiCad version	7.0 or later (8.x recommended: the net-class and DRC-rule syntax in §7 assumes 7+)
Project name	<code>sv16_board</code> — one project, one schematic, five sheets matching <code>BOARD.md</code> §7.1–§7.5
Sheet names	1 power · 2 configuration/JTAG/reset · 3 SPI flashes · 4 clock/reset/console · 5 LEDs/expansion/motor
Board size	100 × 100 mm (five boards for ~₹3,000 at JLCPCB); mounting holes 4 × M3 at the corners, 3 mm from the edges
Stack-up	4 layers, 1.6 mm: signal / solid GND / solid PWR / signal . The inner layers carry the 3V3, 1V1 and 2V5 pours and nothing else
Copper	1 oz outer, 1 oz inner
Min track/clearance	0.2 mm / 0.2 mm (the fab's cheap rules); the FPGA fan-out uses 0.25 mm signal traces and 0.2/0.4 mm vias

Step	Setting
Sheet size	A3 per sheet (this design has 10 connectors and a 144-pin part; A4 gets cramped)
Net naming	use the LPF signal names (clk_25m, uart_rx, flash_cs_n, gpio_a[0] ...) so a net in KiCad can be traced to a line in constraints/ ecp5_144tqfp.lpf and to a row in §5. Power nets: VM_IN, 3V3, 2V5, 1V1, GND
Symbol libraries	Device (R, C, L, D, Q, ferrite bead), Connector_Generic, Connector, Power_Protection (USBLC6), Regulator_Linear (AMS1117), Regulator_Switching (LM2596S-3.3), Memory_Flash (W25Q64JVSSIQ), Interface_USB/Interface_UART — for the CH340C, check the symbol's pin numbers against the WCH datasheet (see §4.7); for the ECP5, Lattice library or draw a 144-pin symbol from board/TQFP144_PINOUT.md
Net classes	§7 defines five: Default, Power, Switching, USB, JTAG

2. KiCad footprint map

Ref	Part	KiCad footprint (library:name)	Notes
U1	LFE5U-12F-6TG144C	Package_QFP:LQFP-144_20x20mm_P0.5mm	0.5 mm pitch, no exposed pad. Verify the courtyard against the datasheet's 22 × 22 mm body
U2, U3	W25Q64JVSSIQ	Package_S0:SOIC-8_5.23x5.23mm_P1.27mm	208 mil body
U4	CH340G	Package_S0:SOIC-16_3.9x9.9mm_P1.27mm	150 mil body. Needs Y2 + C36/C37 (below)
U5	LM2596S-3.3	Package_T0_S0T_SMD:T0-263-5_TabPin3	tab = pin 3 (GND); the tab is the heatsink, pour copper under it
U6	MP1584EN-LF-Z	Package_S0:SOIC-8-1EP_3.9x4.9mm_P1.27mm ⚠	Exposed pad. GND. Check the pad size against the MP1584 datasheet — some SOIC-8E parts use a 2.4 × 3.3 mm pad, others 3.3 × 2.4 mm
U7	AMS1117-2.5	Package_T0_S0T_SMD:S0T-223-3_TabPin3	tab = pin 2 (VOUT)
U8	USBLC6-2SC6	Package_T0_S0T_SMD:S0T-23-6	
Y1	active 25 MHz XO (4-pad: VCC, GND, OUT, EN)	Oscillator:Oscillator_SMD_Abracon_ASV-4Pin_7.0x5.1mm ⚠	5 × 7 mm type Oscillator_SMD_Abracon_ASV-4Pin_7.0x5.1mm ⚠ a 2-pin HC49/US crystal does not fit this and cannot clock the FPGA (§10 correction 5)
Y2	12 MHz crystal (CH340G only)	Crystal:Crystal_HC49-4H_Vertical (through-hole, 4.88 mm lead spacing) or Crystal:Crystal_SMD_3225-4Pin_3.2x2.2mm (2-pin SMD)	must be 12 MHz — XI accepts 11.98–12.02 MHz only, so no other value works. Pick the one you buy; omit Y2, C36, C37 with a CH340C
Q1, Q2, Q3, Q4	2N7002 / BC817 / AO3401 / BC807	Package_T0_S0T_SMD:S0T-23	pin 1 = gate/base on all four

Ref	Part	KiCad footprint (library:name)	Notes
D1-D7	LED 0603	LED_SMD:LED_0603_1608Metric	cathode on the marked end — orientation matters (\$4.9)
D8-D11	SS34 (SMA)	Diode_SMD:D_SMA	pin 1 = cathode (band)
R1-R42	0603 resistors	Resistor_SMD:R_0603_1608Metric	1 %, 0.1 W
C1-C20, C26-C29, C31-C33 + the 100 nF companions	0603 ceramic	Capacitor_SMD:C_0603_1608Metric	X7R, 16 V or 25 V
C25, C30	0805 ceramic	Capacitor_SMD:C_0805_2012Metric	10 µF: 25 V X7R (a 10 µF/16 V 0805 is fine on 3V3, marginal on VM_IN)
C21	100 µF / 10 V electrolytic	Capacitor_SMD:CP_Elec_6.3x5.4 ⚠	verify against the can you buy; 6.3 × 5.4 mm is the common 100 µF/10 V size
C22	22 µF / 10 V electrolytic	Capacitor_SMD:CP_Elec_5x5.4 ⚠	
C23	470 µF / 35 V electrolytic	Capacitor_SMD:CP_Elec_10x10.2 ⚠	10 mm can; 35 V minimum , this sits on VM_IN
C24	220 µF / 16 V electrolytic	Capacitor_SMD:CP_Elec_8x6.9 ⚠	
L1	33 µH / 3 A shielded	Inductor_SMD:L_Bourns_SRR1260 or generic 8 × 8 mm ⚠	pick from the inductor you buy; keep the footprint's pad spacing
L2	10 µH / 3 A shielded	generic 6 × 6 mm shielded ⚠	same
FB1	ferrite bead 0603	Inductor_SMD:L_0603_1608Metric	600 Ω @ 100 MHz, ≥ 500 mA
SW1, SW2	6 × 6 mm tactile	Button_Switch_THT:SW_PUSH_6mm	through-hole for strength; fits the 6 × 6 mm body
J1, J5, J6, J10	1×10 / 1×6 / 1×4 headers	Connector_PinHeader_2.54mm:PinHeader_1x10_P2.54mm_Vertical	if the keyed header is not available, add the keying notch or use a shrouded header
J3, J4	2×10 headers	Connector_PinHeader_2.54mm:PinHeader_2x10_P2.54mm_Vertical	
J7	2×25 header	Connector_PinHeader_2.54mm:PinHeader_2x25_P2.54mm_Vertical	footprint only, not fitted
J2	2×5 box header, 1.27 mm	Connector_PinHeader_1.27mm:PinHeader_2x5_P1.27mm_Vertical	if the connector is unavailable in India, leave J2 unpopulated and use J1
J8	USB-C receptacle 16-pin	Connector_USB:USB_C_Receptacle_HR01 ⚠	the USB-C 16-pin HRO part; if you buy a different receptacle, take the footprint from its datasheet. Micro-USB alternative: Connector_USB:USB_Micro-B_Molex-105017-0001 (omit R36/R37)
J9	5.5/2.1 mm barrel jack	Connector_BarrelJack:BarrelJack_CUI_P102A_Ver1 ⚠	centre positive
JP1	2-pin header + shunt	Connector_PinHeader_2.54mm:PinHeader_1x02_P2.54mm_Vertical	
TP1-TP6	1 mm test pads	TestPoint:TestPoint_Pad_D1.0mm	nets: 3V3, 2V5, 1V1, GND, DONE, INITN

3. Component specifications

3.1 Active devices

Ref	Part	Electrical specification	Package	Qty	Price (India)
U1	LFE5U-12F-6TG144	CECP5 FPGA: 12,144 LUT4, 72 KB sysMEM, 28 × 18-bit multipliers, 2 PLLs. VCC 1.045–1.155 V, VCCAUX 2.375–2.625 V, VCCIO 3.3 V. 98 I/O on this package, 52 used. Speed grade 6, 0–85 °C Tj	TQFP-144 20 × 20 mm, 0.5 mm pitch	1	digkey.in ≈ ₹1,650 — no substitute exists ; see BOARD.md §3.4 for the speed-7 / industrial-grade alternatives
U2	W25Q64JVSSIQ	64 Mbit (8 MB) SPI NOR, 2.7–3.6 V, 104 MHz, standard + dual/quad SPI. Holds the bitstream	SOIC-8, 208 mil	1	Sharvie ₹75 · Hubtronics ₹117
U3	W25Q64JVSSIQ	identical part; holds firmware images (slot A at 0x000000, slot B at 0x008000 — 32 KB each). A W25Q32/W25Q16 is already enough	SOIC-8, 208 mil	1	as U2
U4	CH340G	USB 2.0 full-speed ↔ UART bridge, 3.3 V operation (V3 tied to VCC at 3.3 V), up to 2 Mbps. Requires a 12 MHz crystal (Y2) + 22 pF load caps (C36/C37) — the G has no internal clock. Pin 13 is DTR# (used for auto-reconfigure); R232 (15) has an internal pull-down; CTS/DSR/RI/DCD/RTS are unused	SOP-16, 150 mil	1	iFutureTech · Hubtronics · Robu
Y2	12 MHz crystal	12 MHz quartz, CL 12–20 pF, ESR ≤ 60 Ω, 2-pin passive — oscillated by the CH340G's internal inverter	HC49/US or 3225 SMD	1	Robu / Zbotic ₹15–25
U5	LM2596S-3.3	buck, 3.3 V fixed , VIN 4.5–40 V, 3 A, 150 kHz, non-synchronous, ON/OFF < 1.3 V = on	TO-263-5 (tab = GND)	1	Robu ₹51 (XBLW) / ₹61 (Sikor)

Ref	Part	Electrical specification	Package	Qty	Price (India)
U6	MP1584EN-LF-Z	buck, adjustable 0.8–20 V, VIN 4.5–28 V, 3 A, 100 kHz–1.5 MHz, 0.8 V reference, non-synchronous, EN threshold 1.2 V falling / 1.5 V rising, 1.5 μ A internal EN pull-up	SOIC-8E (EP = GND)	1	Hubtronics ₹47 · Sunrom
U7	AMS1117-2.5	LDO, 2.5 V fixed, 1 A, dropout $\approx$ 1.3 V, requires $\geq$ 10 μ F at the output for stability	SOT-223 (tab = VOUT)	1	Robu / QuartzComponents ₹12–20
U8	USBLC6-2SC6	USB ESD protection: 2 data-line channels + VBUS clamp, 6 V standoff, < 1 nF line capacitance	SOT-23-6	1	Sunrom ₹30 · DNA Tech ₹32
Y1	YIC OSC25M-3.3I/S3-25T	25.000 MHz XO, 3.3 V CMOS, $\pm$ 25 ppm (UART tolerance needs $\pm$ 2 %), enable pin, 15 pF drive	3.2 $\times$ 2.5 mm 4-pad	1	digikey.in ₹111 · 5 $\times$ 7 mm equivalents on Amazon.in/eBay.in
Q1	2N7002	N-MOSFET 60 V / 115 mA, Vgs(th) 2.5 V max — pulls PROGRAMN low from the host's DTR	SOT-23	1	Robu ₹5
Q2	BC817	NPN 45 V / 500 mA — drives the DONE LED	SOT-23	1	₹2
Q3	AO3401	P-MOSFET –30 V / –4 A, Rds(on) 60 m Ω — load switch that delays the 2.5 V LDO	SOT-23	1	₹8
Q4	BC807	PNP 45 V / 500 mA — drives the INITN LED	SOT-23	1	₹2
D1–D5	LED green 0603	Vf $\approx$ 2.0 V @ 5 mA, 20 mA max	0603	5	₹1–2
D6, D7	LED red 0603	Vf $\approx$ 1.9 V; D6 is DNP by default	0603	2	₹1–2
D8, D9	SS34	Schottky 40 V / 3 A, Vf $\approx$ 0.5 V — input ORing between the barrel jack and USB	SMA	2	₹3

Ref	Part	Electrical specification	Package	Qty	Price (India)
D10	SS34	catch diode for the 3V3 LM2596 (obligatory, non-synchronous)	SMA	1	₹3
D11	SS34	catch diode for the 1V1 MP1584	SMA	1	₹3
—	SN74LVC1T45 (optional)	level shifter for a 5 V adapter on J10	SOT-23-6	0	only if needed

3.2 Resistors (all 0603, 1 %, 0.1 W unless noted)

Ref	Value	Connects	Purpose
R1	10 kΩ	ext_rst_n → 3V3	reset pull-up (the pin also has PULLMODE=UP in the LPF)
R2	4.7 kΩ	PROGRAMN → 3V3	configuration pull-up (TN-02038 Table 6.2)
R3, R4	4.7 kΩ	INITN → 3V3, DONE → 3V3	status pull-ups
R5	4.7 kΩ	CFG_1 (pin 59) → 3V3	Master-SPI mode strap, CFG[2:0] = 010
R6, R7	1 kΩ	CFG_0 (62) → GND, CFG_2 (58) → GND	mode straps
R8	1 kΩ	CCLK (54) → 3V3	MCLK pull-up
R9-R12	100 Ω	the four Master-SPI nets, at the FPGA end	damping; bounds contention with GPIOA[1]/[2]/[4]/[6]
R13-R15	10 kΩ	U2 /CS, /WP, /HOLD → 3V3	flash deselect/idle
R16-R18	10 kΩ	U3 /CS, /WP, /HOLD → 3V3	flash deselect/idle
R19	10 kΩ	motor_fault_n → 3V3	motor fault pull-up
R21, R23	0 Ω	CH340C TXD → uart_rx, RXD → uart_tx	UART routing option A
R22, R24	0 Ω	J10 → uart_rx / uart_tx	UART routing option B — populate one pair only
R25	4.7 kΩ	Q1 gate series	gate damping
R26	10 kΩ	Q1 gate → GND	keeps PROGRAMN released when DTR floats
R27-R30	470 Ω	led[0..3] → D2-D5 → 3V3	~4 mA per LED
R31	1 kΩ	INITN → Q4 base	Q4 drive
R32	10 kΩ	DONE → Q2 base	Q2 drive
R33	1 kΩ	Q2 collector → D1 → 3V3	D1 current limit
R34	1 kΩ	Q4 collector → D6 → GND	D6 current limit
R36, R37	5.1 kΩ	USB-C CC1, CC2 → GND	Rd for a UFP power sink (omit with micro-USB)
R38	12.4 kΩ (E96)	1V1 → U6 FB	feedback divider top — see §10 correction 2

Ref	Value	Connects	Purpose
R39	33 kΩ	U6 FB → GND	feedback divider bottom — must stay below 40 kΩ (§10 correction 2)
R40	100 kΩ	U6 FREQ → GND	switching frequency: the MP1584 datasheet's table gives 900 kHz at 100 kΩ (formula $R = 180000 / f^{1.1}$, R in kΩ, f in kHz)
R41	100 kΩ	VM_IN → U6 EN, with C31 to GND	sequencing delay, $\tau = 47$ ms
R42	1 MΩ	Q3 gate → GND	turns Q3 on; with C32 this is the 2V5 delay (§10 correction 4)
R43	100 kΩ	Q3 gate → Q3 source	Vgs clamp: -10.9 V at VM_IN = 12 V, inside the ± 12 V rating (§10 correction 4)

3.3 Capacitors

Ref	Value	Package	Purpose
C1	100 nF	0603	reset RC with R1 (noise and bounce, not timing)
C2-C7	100 nF ×6	0603	VCC decoupling, one per pin: 20, 29, 38, 66, 83, 130
C8-C11	100 nF ×4	0603	VCCAUX decoupling: pins 17, 53, 96, 132
C12-C20	100 nF ×9	0603	VCCIO decoupling: pins 9, 16, 36, 43, 70, 86, 100, 122, 137
C21	100 μF / 10 V electrolytic + 100 nF	6.3 × 5.4 + 0603	1V1 bulk
C22	22 μF / 10 V electrolytic + 100 nF	5 × 5 + 0603	2V5 bulk
C23	470 μF / 35 V electrolytic + 100 nF	10 × 10 + 0603	LM2596 input bulk (datasheet value for this input range)
C24	220 μF / 16 V electrolytic + 100 nF	8 × 6.9 + 0603	LM2596 output bulk on 3V3
C25	10 μF	0805	3V3 bulk near banks 0/1/6/7
C26	100 nF	0603	Y1 decoupling
C27	100 nF	0603	CH340C decoupling (plus the 100 nF on V3)
C28, C29	100 nF	0603	U2, U3 decoupling
C30	10 μF in + 10 μF out	0805 ×2	AMS1117 input/output — it needs ≥ 10 μF; bench-check for ringing (OQ-B10)
C31	470 nF	0603	U6 EN delay with R41 (47 ms time constant)

Ref	Value	Package	Purpose
C32	4.7 μF	0805	Q3 gate→ source delay with R42/R43: $\tau = 91 \text{ k}\Omega \times 4.7 \text{ }\mu\text{F} \approx 428 \text{ ms} \rightarrow \approx 54 \text{ ms}$ to threshold at 12 V
C33	100 nF	0603	U6 bootstrap, BST → SW

3.4 Inductors and filtering

Ref	Value	Specification	Package	Purpose
L1	33 μ H	$\geq 3 \text{ A}$ saturation, shielded, low DCR	8 × 8 mm	LM2596 3V3 inductor (datasheet value for 3.3 V fixed)
L2	10 μ H	$\geq 3 \text{ A}$ saturation, shielded	6 × 6 mm	MP1584 1V1 inductor
FB1	600 Ω @ 100 MHz	$\geq 500 \text{ mA}$, 0603	0603	VM_IN filtering ahead of the regulators

3.5 Connectors, switches and test points

Ref	Item	Pinout (by pin number)	Notes
J1	1×10 header, 2.54 mm	1 3V3 · 2 TDO · 3 TDI · 4 PROGRAMn · 5 NC · 6 TMS · 7 GND · 8 TCK · 9 DONE · 10 INITn	Versa-style; works with openFPGALoader and FT232H/FT2232H adapters
J2	2×5 box header, 1.27 mm	1 3V3 · 2 TMS · 3 TDI · 4 TDO · 5 TCK · 6 GND · 7–9 NC · 10 GND	wired in parallel with J1 — use one at a time
J3	2×10, 2.54 mm	1 3V3 · 2 GND · 3 A8 · 4 A9 · 5 A10 · 6 A11 · 7 A12 · 8 A13 · 9 A14 · 10 A15 · 11 spi0_cs_n · 12 spi0_sck · 13 spi0_mosi · 14 spi0_miso · 15,16 GND · 17,18 5V · 19 GND · 20 GND	EXP-A: SPI0 + GPIOA[15:8]. The 5 V pins are USB VBUS only
J4	2×10, 2.54 mm	1 3V3 · 2 GND · 3–18 B0–B15 · 19 GND · 20 5V	EXP-B: all 16 GPIOB
J5	1×10, 2.54 mm	1 A0 · 2 A1 · 3 A2 · 4 A3 · 5 A4 · 6 A5 · 7 A6 · 8 A7 · 9 GND · 10 3V3	EXP-C. Pins 2/3/5/7 are also the configuration-flash nets — inputs only (BOARD.md §6.4)
J6	1×6 keyed	1 VM (VM_IN) · 2 GND · 3 pwm_out · 4 motor_dir1 · 5 motor_dir2 · 6 motor_fault_n	motor driver interface; the fuse lives on the driver board
J7	2×25, 2.54 mm	46 spare I/O + 4 GND	footprint only, not fitted
J8	USB-C receptacle 16 pin	VBUS, GND, D+, D–, CC1, CC2 (Rd = 5.1 k Ω)	power + console on one cable
J9	5.5/2.1 mm barrel	centre positive, 7–12 V	bench supply; see BOARD.md OQ-B12 for the 5 V USB-load caveat
J10	1×4 header	1 3V3 · 2 uart_tx · 3 uart_rx · 4 GND	console without the CH340C (fit R22/R24, remove R21/R23)
SW1	6 × 6 mm tactile	ext_rst_n → GND	reset

Ref	Item	Pinout (by pin number)	Notes
SW2	6 × 6 mm tactile	PROGRAMN → GND	reconfigure from flash
JP1	2-pin + shunt	in series with U4 DTR# → Q1 gate	remove to disable auto-reconfigure
TP1-TP6	1 mm pads	3V3, 2V5, 1V1, GND, DONE, INITN	top side, labelled with these net names

4. Pin-level connections

Everything in this section comes from the five connection sheets in `BOARD.md` §7. Pin names are what matter; where a part's pin **numbers** depend on the vendor's drawing, the row says "confirm".

4.1 U1 — the FPGA (power, configuration and JTAG)

U1 pin	Net / connection
20, 29, 38, 66, 83, 130	1V1 (+ C2-C7, 100 nF each)
17, 53, 96, 132	2V5 (+ C8-C11, 100 nF each)
9, 16, 36, 43, 70, 86, 100, 122, 137	3V3 (+ C12-C20, 100 nF each)
8, 15, 21, 32, 42, 65, 75, 85, 87, 101, 123, 129, 131, 138	GND — each with its own via to the ground plane
109, 144	not connected
54	CCLK → R9 100 Ω → U2.6, and R8 1 kΩ → 3V3
46	Master-SPI D1/MISO pad → R11 100 Ω → U2.2 (D0); also gpio_a[1] on J5.2
47	Master-SPI D0/MOSI pad → R10 100 Ω → U2.5 (DI); also gpio_a[2] on J5.3
49	CSSPIN pad → R12 100 Ω → U2.1 (/CS); also gpio_a[4] on J5.5
51	Master-SPI DOUT/CS0N pad — unused by the design; also gpio_a[6] on J5.7
57	PROGRAMN → R2 4.7 kΩ → 3V3; SW2 → GND; J1.4; Q1 drain
55	INITN → R3 4.7 kΩ → 3V3; J1.10; R31 1 kΩ → Q4 base
56	DONE → R4 4.7 kΩ → 3V3; J1.9; R32 10 kΩ → Q2 base
62, 59, 58	CFG_0 → R6 1 kΩ → GND · CFG_1 → R5 4.7 kΩ → 3V3 · CFG_2 → R7 1 kΩ → GND
63, 64, 61, 60	TCK, TMS, TDI, TDO → J1.8/6/3/2 and J2.5/2/3/4
133	clk_25m ← Y1.3 (25 MHz oscillator output)
134	ext_rst_n → SW1, R1 10 kΩ → 3V3, C1 100 nF → GND
73, 74	uart_rx, uart_tx → U4 or J10 (through R21-R24)
110, 111, 112, 113	flash_sck, flash_cs_n, flash_mosi, flash_miso → U3.6, U3.1, U3.5, U3.2
114-117	spi0_sck, spi0_cs_n, spi0_mosi, spi0_miso → J3.12/11/13/14
39, 40, 41, 44	led[0..3] → R27-R30 470 Ω → D2-D5 → 3V3 (active low)
45-52	gpio_a[0..7] → J5.1-J5.8 (pins 2/3/5/7 also reach the config flash)

U1 pin	Net / connection
97-99, 104-108	gpio_a[8..15] → J3.3-J3.10
135, 136, 139-143, 128, 124-127, 1-4	gpio_b[0..15] → J4.3-J4.18
88, 89, 102, 103	pwm_out, motor_dir1, motor_dir2, motor_fault_n → J6.3-J6.6

The complete 144-pin table, including the 46 unassigned I/O pins and every pad name, is board/TQFP144_PINOUT.md (generated by scripts/sv16_board_pins.py). §5 below is the subset the design actually constrains, generated from the LPF.

4.2 U2 — configuration flash (write-protected, driven by the FPGA)

U2 pin	Net
1 /CS	R12 100 Ω → U1.49, and R13 10 k Ω → 3V3
2 D0	R11 100 Ω → U1.46
3 /WP	R14 10 k Ω → 3V3
4 GND	GND
5 DI	R10 100 Ω → U1.47
6 CLK	R9 100 Ω → U1.54
7 /HOLD	R15 10 k Ω → 3V3 (IO2/IO3 unused: no quad SPI on TQFP-144)
8 VCC	3V3 + C28 100 nF

4.3 U3 — application flash (firmware, driven by the soft-core)

U3 pin	Net
1 /CS	U1.111 flash_cs_n, and R16 10 k Ω → 3V3
2 D0	U1.113 flash_miso
3 /WP	R17 10 k Ω → 3V3
4 GND	GND
5 DI	U1.112 flash_mosi
6 CLK	U1.110 flash_sck
7 /HOLD	R18 10 k Ω → 3V3
8 VCC	3V3 + C29 100 nF

4.4 U4 — CH340G USB-UART bridge (pin numbers from the WCH datasheet)

U4 pin	Name	Net
1	GND	GND
2	TXD	R21 0 Ω → U1.73 uart_rx (idles high)
3	RXD	R23 0 Ω → U1.74 uart_tx
4	V3	3V3 + 100 nF (3.3 V operation ties V3 to VCC)
5	UD+	J8 D+ through U8 (D+ channel)
6	UD-	J8 D- through U8 (D- channel)

U4 pin	Name	Net
7	XI	Y2.1, C36 22 pF → GND (or an active 12 MHz oscillator, with X0 left open)
8	X0	Y2.2, C37 22 pF → GND (leave open when XI is driven by an oscillator)
9, 10, 11, 12, 14	CTS#, DSR#, RI#, DCD#, RTS#	not connected
13	DTR#	JP1 → Q1 gate (auto-reconfigure; remove JP1 to disable)
15	R232	not connected (internal pull-down)
16	VCC	3V3 + C27 100 nF

Keep the XI/XO traces short and symmetrical, and away from the USB pair and the switchers.

Pick **C36/C37 to suit the crystal's load capacitance (CL)**: $C = 2 \times (CL - 3...5 \text{ pF stray})$, so a **CL = 12 pF** crystal wants 18–22 pF and the common **CL = 20 pF** HC49/US type wants 33 pF. WCH's own reference schematic quotes 22 pF in one datasheet revision and 33 pF in another for exactly this reason. Getting it wrong does not stop the chip working — it pulls the frequency a few tens of ppm, well inside the 11.98–12.02 MHz window — but matching is free, so read the crystal's label.

The two-pin crystal works here because the CH340G has its **own inverter** on XI/X0 (unlike the ECP5, which has no crystal driver — that is why Y1 must be an active part, §10 correction 5). If you would rather use an active part, the same datasheet says: "When using an external oscillator feed the clock signal into XI pin, and leave XO pin unconnected." An active 12 MHz 3.3 V XO into XI with X0 open also works — it simply costs more than a ₹20 crystal.

4.5 U5 — LM2596S-3.3 (3V3 rail, from VM_IN)

U5 pin	Name	Connection
1	VIN	VM_IN after FB1, with C23 470 µF and 100 nF to GND
2	OUT	switch node: L1 33 µH → 3V3; D10 cathode here (anode → GND)
3	GND (tab)	GND — copper pour under the tab
4	FB	3V3 — the fixed 3.3 V version senses its own output here (TI SNVS124: the FB pin is "removed from output" only in the force-on/force-off test procedures)
5	ON/OFF	GND (always enabled; < 1.3 V = on)

Output side: C24 220 µF + 100 nF on 3V3.

4.6 U6 — MP1584EN (1V1 rail, from VM_IN)

U6 pin	Name	Connection
1	SW	L2 10 µH → 1V1; D11 cathode here (anode → GND); C33 100 nF to BST
2	EN	VM_IN → R41 100 kΩ → EN, C31 470 nF EN → GND ($\tau = 47 \text{ ms}$; crosses the 1.5 V rising threshold in $\approx 6 \text{ ms}$ from 12 V, $\approx 13 \text{ ms}$ from 5 V)

U6 pin	Name	Connection
3	COMP	⚠ not yet specified — the part needs an RC compensation network to GND (datasheet typical application, values depend on Vout and fsw). Add it before layout: §10 correction 3
4	FB	divider from 1V1: R38 top, R39 bottom — see §10 correction 2 for the recommended values
5	GND	GND (exposed pad to the ground plane, vias)
6	FREQ	R40 100 kΩ → GND ⇒ 900 kHz
7	VIN	VM_IN with 100 nF
8	BST	C33 100 nF → SW

Output side: C21 100 μF + 100 nF on 1V1. **Measure this rail before fitting U1** (BOARD.md §10 step 2).

4.7 U7 and Q3 — the delayed 2V5 rail

Device	Pin	Connection
Q3 (AO3401)	S	VM_IN (the AMS1117 needs ≥ 3.8 V in for 2.5 V out, so this section cannot run from 3V3)
	G	R42 1 MΩ → GND, C32 4.7 μF → S (source), R43 100 kΩ → S as the Vgs clamp
	D	U7 VIN
U7 (AMS1117-2.5)	1 GND	GND
	2 VOUT (tab)	2V5 + C22 22 μF + 100 nF
	3 VIN	Q3 drain + C30 10 μF to GND

4.8 U8 — USB ESD protection ⚠ (confirm pin numbers against the ST datasheet)

U8 pin	Connection
1 I/01	J8 D+ (connector side)
2 GND	GND
3 I/02	J8 D−
4 I/02'	U4 UD− (protected side)
5 VBUS	J8 VBUS
6 I/01'	U4 UD+

Place U8 at the connector; the D+/D− pair runs from J8 through U8 to U4 as a 90 Ω differential pair. CC1/CC2 → R36/R37 5.1 kΩ → GND, within 10 mm of J8.

4.9 Status and user LEDs (all active low)

Net	Chain
led[0] (U1.39)	R27 470 Ω → D2 → 3V3

Net	Chain
led[1] (U1.40)	R28 470 Ω → D3 → 3V3
led[2] (U1.41)	R29 470 Ω → D4 → 3V3
led[3] (U1.44)	R30 470 Ω → D5 → 3V3
DONE (U1.56)	R32 10 k Ω → Q2 base; Q2 emitter → GND; Q2 collector → D1 → R33 1 k Ω → 3V3 (D1 lights when DONE is high)
INITN (U1.55)	R31 1 k Ω → Q4 base; Q4 emitter → 3V3; Q4 collector → D6 → R34 1 k Ω → GND (D6 lights when INITN is low; D6 is DNP by default)
3V3 present	D7 + its series resistor from 3V3 to GND

5. FPGA pin table — generated from the constraint file

The table below is generated by `scripts/sv16_pcb_doc.py` directly from `constraints/ecp5_144tqfp.lpf`, so it cannot drift from the bitstream's pin map. `make lint` fails if it is stale.

TQFP pin	Bank	Port (LPF name)	IO type	Drive	Pull	Board net
1	7	gpio_b[12]	LVC MOS33	8 mA	NONE	J4.15 (EXP-B)
2	7	gpio_b[13]	LVC MOS33	8 mA	NONE	J4.16 (EXP-B)
3	7	gpio_b[14]	LVC MOS33	8 mA	NONE	J4.17 (EXP-B)
4	7	gpio_b[15]	LVC MOS33	8 mA	NONE	J4.18 (EXP-B)
39	8	led[0]	LVC MOS33	8 mA	NONE	D2 + R27 (status LED, active low)
40	8	led[1]	LVC MOS33	8 mA	NONE	D3 + R28 (status LED, active low)
41	8	led[2]	LVC MOS33	8 mA	NONE	D4 + R29 (status LED, active low)
44	8	led[3]	LVC MOS33	8 mA	NONE	D5 + R30 (status LED, active low)
45	8	gpio_a[0]	LVC MOS33	8 mA	NONE	J5.1 (EXP-C)
46	8	gpio_a[1]	LVC MOS33	8 mA	NONE	J5.2 (EXP-C) + U2 config-flash net (BOARD.md 6.4)
47	8	gpio_a[2]	LVC MOS33	8 mA	NONE	J5.3 (EXP-C) + U2 config-flash net (BOARD.md 6.4)
48	8	gpio_a[3]	LVC MOS33	8 mA	NONE	J5.4 (EXP-C)
49	8	gpio_a[4]	LVC MOS33	8 mA	NONE	J5.5 (EXP-C) + U2 config-flash net (BOARD.md 6.4)

TQFP pin	Bank	Port (LPF name)	IO type	Drive	Pull	Board net
50	8	gpio_a[5]	LVC MOS33	8 mA	NONE	J5.6 (EXP-C)
51	8	gpio_a[6]	LVC MOS33	8 mA	NONE	J5.7 (EXP-C) + U2 config-flash net (BOARD.md 6.4)
52	8	gpio_a[7]	LVC MOS33	8 mA	NONE	J5.8 (EXP-C)
73	3	uart_rx	LVC MOS33	4 mA	UP	U4 TXD via R21 (or J10.3 via R22)
74	3	uart_tx	LVC MOS33	4 mA	NONE	U4 RXD via R23 (or J10.2 via R24)
88	2	pwm_out	LVC MOS33	16 mA	NONE	J6.3 (motor PWM)
89	2	motor_dir1	LVC MOS33	8 mA	NONE	J6.4 (motor DIR1)
97	2	gpio_a[8]	LVC MOS33	8 mA	NONE	J3.3 (EXP-A)
98	2	gpio_a[9]	LVC MOS33	8 mA	NONE	J3.4 (EXP-A)
99	2	gpio_a[10]	LVC MOS33	8 mA	NONE	J3.5 (EXP-A)
102	2	motor_dir2	LVC MOS33	8 mA	NONE	J6.5 (motor DIR2)
103	2	motor_fault_n	LVC MOS33	4 mA	UP	J6.6 (motor FAULTn) + R19 pull-up
104	2	gpio_a[11]	LVC MOS33	8 mA	NONE	J3.6 (EXP-A)
105	2	gpio_a[12]	LVC MOS33	8 mA	NONE	J3.7 (EXP-A)
106	2	gpio_a[13]	LVC MOS33	8 mA	NONE	J3.8 (EXP-A)
107	2	gpio_a[14]	LVC MOS33	8 mA	NONE	J3.9 (EXP-A)
108	2	gpio_a[15]	LVC MOS33	8 mA	NONE	J3.10 (EXP-A)
110	1	flash_sck	LVC MOS33	8 mA	NONE	U3.6 CLK (application flash)
111	1	flash_cs_n	LVC MOS33	8 mA	NONE	U3.1 /CS
112	1	flash_mosi	LVC MOS33	8 mA	NONE	U3.5 DI
113	1	flash_miso	LVC MOS33	4 mA	UP	U3.2 DO
114	1	spi0_sck	LVC MOS33	8 mA	NONE	J3.12 (EXP-A)
115	1	spi0_cs_n	LVC MOS33	8 mA	NONE	J3.11 (EXP-A)
116	1	spi0_mosi	LVC MOS33	8 mA	NONE	J3.13 (EXP-A)
117	1	spi0_miso	LVC MOS33	4 mA	UP	J3.14 (EXP-A)
124	1	gpio_b[8]	LVC MOS33	8 mA	NONE	J4.11 (EXP-B)
125	1	gpio_b[9]	LVC MOS33	8 mA	NONE	J4.12 (EXP-B)

TQFP pin	Bank	Port (LPF name)	IO type	Drive	Pull	Board net
126	1	gpio_b[10]	LVC MOS33	8 mA	NONE	J4.13 (EXP-B)
127	1	gpio_b[11]	LVC MOS33	8 mA	NONE	J4.14 (EXP-B)
128	0	gpio_b[7]	LVC MOS33	8 mA	NONE	J4.10 (EXP-B)
133	0	clk_25m	LVC MOS33	4 mA	NONE	Y1.3 (25 MHz oscillator output)
134	0	ext_rst_n	LVC MOS33	4 mA	UP	SW1 + R1 + C1 (reset button)
135	0	gpio_b[0]	LVC MOS33	8 mA	NONE	J4.3 (EXP-B)
136	0	gpio_b[1]	LVC MOS33	8 mA	NONE	J4.4 (EXP-B)
139	0	gpio_b[2]	LVC MOS33	8 mA	NONE	J4.5 (EXP-B)
140	0	gpio_b[3]	LVC MOS33	8 mA	NONE	J4.6 (EXP-B)
141	0	gpio_b[4]	LVC MOS33	8 mA	NONE	J4.7 (EXP-B)
142	0	gpio_b[5]	LVC MOS33	8 mA	NONE	J4.8 (EXP-B)
143	0	gpio_b[6]	LVC MOS33	8 mA	NONE	J4.9 (EXP-B)

52 constrained ports. Bank numbers come from `board/ecp5_tqfp144.json`; every bank on this board is 3V3 (VCCIO8 = 3V3 for the configuration and JTAG domains).

The remaining 46 I/O pins are unassigned spares (bank 1: 118–121; bank 2: 90–95; bank 3: 67–69, 71, 72, 76–82, 84; bank 6: 18, 19, 22–28, 30, 31, 33–35, 37; bank 7: 5–7, 10–14) — they go to J7 and nothing else. `board/TQFP144_PINOUT.md` lists all 144 pins with pad names and banks.

6. Net classes for the schematic

Net class	Nets	Why it exists
Power	VM_IN, 3V3, 1V1, 2V5	wide traces, the pours, and the regulator loops
Switching	U5.OUT/L1 node, U6.SW, BST	short, fat, away from the oscillator and flash; this is where EMI comes from
USB	USB_D+, USB_D–	90 Ω differential pair, ESD device at the connector
JTAG	TCK, TMS, TDI, TDO, PROGRAMN, INITN, DONE, CCLK	short stubs, series resistors at the FPGA end
Default	everything else	0.25 mm signal

Net names to use: the LPF signal names for FPGA pins (`clk_25m`, `ext_rst_n`, `uart_rx`, `uart_tx`, `flash_sck`, `flash_cs_n`, `flash_mosi`, `flash_miso`, `spi0_sck`, `spi0_cs_n`, `spi0_mosi`, `spi0_miso`, `led[0..3]`, `gpio_a[0..15]`, `gpio_b[0..15]`, `pwm_out`, `motor_dir1`, `motor_dir2`, `motor_fault_n`) and `VM_IN` / `3V3` / `2V5` / `1V1` / `GND` for power. Nothing else — a net that exists only in KiCad cannot be checked against the design.

7. Layout rules as KiCad DRC settings

Rule	Setting in KiCad
Track width, signal	0.25 mm (Default net class)
Track width, power	0.8 mm for 3V3, 1V1, 2V5; 1.5 mm for VM_IN — or use the PWR plane for the three rails
Track width, switching nodes	0.8 mm, as short as possible: U5.OUT→L1, U6.SW→L2, and the catch-diode loops
Differential pair	USB_D+/USB_D-: 90 Ω, 0.2 mm gap, equal length within 0.5 mm
Vias	0.2 mm drill / 0.4 mm pad (0.3/0.6 for power)
Clearance	0.2 mm minimum; 0.5 mm between Switching nets and everything in the JTAG/flash group
FPGA decoupling	C2–C20 within 3 mm of the matching pin, on the same side, via first at the pad
FPGA ground	every GND pin gets its own via to the inner GND plane; thermal vias under the body
Oscillator	Y1 within 10 mm of pin 133, no via in the trace, ground pour under it, ≥ 5 mm from switching nodes
Config flash	U2 within 15 mm of pins 46/47/49/54; R9–R12 at the FPGA end
Motor return	J6 GND returns to the C23 area on its own path — never through the FPGA ground
Keep-out	no copper pour under the USB-C shell's shield pads or under the barrel jack body
Test points	TP1–TP6 on the top side, with their net names in silkscreen
Silkscreen	name the headers (J1 JTAG, J3 EXP-A, J4 EXP-B, J5 EXP-C, J6 MOTOR, J10 CONSOLE), the rails at TP1–TP3, "5V only" at J3/J4 pins 17/18, and "CFG FLASH - INPUTS ONLY" at J5 pins 2/3/5/7

8. Assembly notes

Step	Note
Order of fitting	passives → connectors → regulators → U1 last (after the rails are measured, BOARD.md §10 step 2)
QFP-144	stencil + paste + hot air; inspect at 10× before power. Do not try to drag-solder 0.5 mm pitch without flux
TO-263 (U5)	solder the tab down and pour copper into it — that tab is the only heatsink
SOIC-8E (U6)	the exposed pad must be soldered (vias to the ground plane); a part soldered on its pins only will overheat
Polarity	D1–D7 cathode = the marked end; D8–D11 band = cathode (pin 1); electrolytics have a polarity stripe
DNP list (by default)	D6 (INITN LED), J7 (spare-I/O header), the optional level shifter — mark them on the silkscreen so a reviewer does not think they are missing
Note for KiCad	put the DNP parts in the schematic as normal but unticked in the "Exclude from board/BOM" fields, so the BOM the fab sees matches BOARD.md §3

9. Ordering list and cost

Group	What to order	₹ (single board)
FPGA	2 × LFE5U-12F-6TG144C (digikey.in / mouser.in) — one spare	3,300
Tier-1 actives	LM2596S-3.3, MP1584EN, AMS1117-2.5, CH340C, 2 × W25Q64JVSSIQ, USBLC6-2SC6, 2N7002, BC817, BC807, AO3401	350
Oscillator	YIC OSC25M-3.3I/S3-25T	111
Passives	0603 resistor + capacitor assortment kit (covers every R and C on this board), plus 33 µH, 10 µH, ferrite bead, 4 electrolytics, 4 × SS34	750
LEDs	5 green + 2 red 0603	12
Connectors	J1-J10, SW1/SW2, JP1, shunts	130
PCB	5 × 4-layer 100 × 100 mm + stencil	3,000
Total	(excluding the spare FPGA and tools)	≈ ₹4,350

A complete order list with the vendors and the module-first alternative is `BOARD.md` §3.2–§3.5.

10. Accuracy notes — what is fixed, and the four things to correct before layout

Fixed and traceable: every component here is the part finalised in `BOARD.md` §3, the FPGA pin map is generated from `constraints/`

`ecp5_144tqfp.lpf`, and the connections come from the five sheets in `BOARD.md` §7. The transistor and diode pin conventions, the flash pinouts and the oscillator pinout are standard for those packages.

Five corrections found while writing this document — apply them to the schematic (all four are now also recorded in `BOARD.md`):

- U5 pin 4 (FB) must be connected to 3V3.** The earlier sheet shows only VIN, OUT, GND and ON/OFF. For the fixed 3.3 V version the feedback pin senses the output; the TI datasheet's own force-on/force-off test procedures are written as "feedback pin removed from output", which only makes sense if it is normally connected to it. Leaving it floating is not a valid configuration.
- U6 feedback divider: 100 kΩ at the bottom is too high for this part.** The MP1584 datasheet says the FB bias current ($\approx 20 \mu\text{A}$) is why the lower divider resistor should stay **below 40 kΩ**. With $R39 = 100 \text{ k}\Omega$ the divider current is only $\approx 11 \mu\text{A}$, so the bias current shifts the output well above the intended 1.10 V — on a rail whose window is 1.045–1.155 V ($\pm 5 \%$), that is not a margin you can spend. Recommended divider: **R39 = 33 kΩ, R38 = 12.4 kΩ (E96)** $\Rightarrow 0.8 \times (1 + 12.4/33) = 1.1006 \text{ V}$, divider current 24 μA , both resistors inside the datasheet's guidance. (`BOARD.md` §3.3, OQ-B10.)
- U6 COMP pin needs an RC network.** The MP1584 requires loop compensation on COMP (an RC to GND); the board sheets do not show it. Take the values from the datasheet's typical application for your output voltage and switching frequency, add them to sheet 1, and re-check the loop on the bench during bring-up. Without it the regulator will be marginally stable at best.
- The 2V5 load switch was drawn the wrong way round, and its gate drive exceeded the FET's rating.** As originally written (R42 gate→source, C32 gate→GND) the gate sits at source potential through the discharged cap at $t = 0$, so **Q3 is on immediately and turns off later** — the opposite of a delayed turn-on. Corrected topology: **R42 1 MΩ gate→GND** (this is what turns the switch on), **C32 4.7 µF gate→source** (the delay: $\tau = (R42 \parallel R43) \cdot C32 \approx 428 \text{ ms}$, reaching the -1.3 V threshold at $\approx 54 \text{ ms}$ from 12 V), and a new **R43 100 kΩ gate→source** to clamp the gate drive. Without R43 the gate settles at GND and $V_{\text{gs}} = -V_{\text{M_IN}} = -12 \text{ V}$ at the maximum input, right on the AO3401's absolute maximum. Also corrected on

the same sheet: this section runs from **VM_IN**, not 3V3 — an AMS1117-2.5 needs at least ≈ 3.8 V of input for a 2.5 V output, so a 3V3 source could never regulate.

5. **U4 is a CH340G and needs a 12 MHz crystal** (pin 7 **XI** / pin 8 **X0**, plus two 22 pF caps), and **Y1 must be an active oscillator**. Both came out of the cart reconciliation: the CH340G has no internal clock (the CH340C does — it is pin-compatible and needs neither Y2 nor C36/C37), and the HC49/US package is a **2-pin passive crystal**: it needs an oscillator circuit and the ECP5 has no on-chip crystal driver, so a passive part on `clk_25m` would leave the board with no clock at all. Two ways to keep the crystal you bought: * **Buy an active 25 MHz XO** (3.3 V, 4-pin) — recommended, $\sim ₹110$, and it is what the board file already has a footprint for. * **Build a Pierce oscillator** around the crystal: an unbuffered inverter (74LVC1G04, SOT-23-5), 1 M Ω feedback, $2 \times \sim 22$ pF load caps (match the crystal's CL), optional 470 Ω series to the clock pin. It works at 25 MHz but adds three parts, needs a clean layout, and is a bring-up risk — only do it if the active part cannot be sourced.

Rows marked $\triangle$ need your datasheet, not mine: the MP1584 exposed-pad size, the electrolytic can sizes, the inductor footprints, the USB-C receptacle and barrel-jack footprints (they vary by the exact part you buy), and the CH340C and USBLC6 pin numbers — the connections above are by pin name, which is what the datasheet will confirm.

What this document cannot tell you: whether your schematic or layout is correct. That is what the review step in `TEAM_PLAN.md` §13.1 (P1 + P4 review before the PCB order on day 10) is for, and what `BOARD.md` §10's bring-up sequence is for.

Revision 1.0 — written against commit 6011293, from `BOARD.md` revision 1.1 (India-sourced BOM) and `constraints/ecp5_144tqfp.lpf`. Rebuild: `make pcb-doc` (regenerates §5 and re-renders this PDF).